

REQUIREMENTS SPECIFICATION

Restaurant Table Monitoring Vision AI Model

Version	1.0
Name	Kannan S
Reviewer	Shrikanth H
Date	20/04/2026

WG Tech Solutions Pvt Ltd

Introduction

1. Purpose

This document defines the requirements for creating restaurant table monitoring which provides details of vision analytics providing – waiter time to engage and take order, table occupancy details, time for food to arrive, time taken by customer at the table

2. Functional Requirements

Description

Detect and provide analytics on table usage in Commercial Restaurants using Vision Analytics

Functional Requirements

1. The system shall detect all persons in video frames
2. The system shall support multiple persons simultaneously – differentiate the waiter based on uniform

3. Hardware and Details of Functional Requirements

The system should be implemented using Raspberry Pi + Axeleram.2 1GB. If the AI model can be done reliably (with processing all frames and good frame rate and alerts happening within 100ms) using Raspberry Pi for single channel, then no need of Axeleram GPU otherwise we will need to run the AI model on Axeleram M.2 GPU. One of the key requirements is to ensure all aspects of analytics for table – occupancy, table usage timeline, service timeline are all monitored. Camera resolution used should be 2MP (can go to 4MP if distances are greater). Video feed for this will be provided for development of AI model.

4. Key Technologies Used

1. Vision AI Analytics
2. Customized AI Model for different use cases – different lighting conditions; restaurant settings

WG Tech Solutions Pvt Ltd

Rainmaker Workspaces, No. 759, 3rd Floor, 8th Main Rd, 3rd Phase,

J. P. Nagar, Bengaluru, Karnataka 560078

WG Tech Solutions Pvt Ltd

5. Deliverables

Following are the deliverables

1. AI Model for table monitoring in restaurants
2. Hardware for running the AI model – Raspberry Pi with Axelera M.2 will be provided. All video feeds for training will be provided

Proposed Timelines and Resources

2 Months for delivery of complete solution

Resources – 2 interns with Stipend of Rs15000 each/month; 1 resource from WGTech supporting/mentoring. IP will be owned by WGTech for this project.

1. AI Model Development, training and customization – 1month
2. Linux OS and Embedded Alignment – hardware ready for AI model deployment including image processing – 1month
3. Complete AI model, hardware testing in real world environment and improvement of model based on feedback – 1 month

WG Tech Solutions Pvt Ltd

Rainmaker Workspaces, No. 759, 3rd Floor, 8th Main Rd, 3rd Phase,

J. P. Nagar, Bengaluru, Karnataka 560078